

Minji Kim M.S.

I am a Master's student at the Machine Perception and Intelligence Lab (MPIL) under the supervision of Prof. Pyojin Kim.

My research focuses on 3D computer vision and robotics. Currently, I am working on indoor localization in extreme environments using limited sensors, and I am also interested in sensor calibration and multi-modal data fusion.



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📁 Portfolio (Korean)

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🌐 Personal Page

🌐 LinkedIn

Language

Opic (English) 📖 IH Intermediate High
TOEIC (English) 📖 850 / 990

Education

2024.02 – Present 📖 **M.S., Mechanical and Robotics Engineering**
Gwangju Institute of Science and Technology.
2018.02 – 2024.02 📖 **B.S., Mechanical Engineering**
Sookmyung Women's University.

Employment History

2025.03 – 2025.06 📖 **NUS Temasek Laboratories**, National University of Singapore.
2022.10 – 2024.02 📖 **Machine Perception and Intelligence Lab**, Sookmyung Women's University.

Research Publications

- 1 **Minji Kim**, J. Han, J. Ham, and P. Kim, "SPLiCE: Single-point LiDAR and camera calibration & estimation leveraging Manhattan World," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- 2 J. Han, Z. Hu, S. Yang, **Minji Kim**, and P. Kim, "SoMaSLAM: 2d graph SLAM for sparse range sensing with soft Manhattan World constraints," *IEEE Robotics and Automation Letters (RA-L)*, 2025.
- 3 J. Ham, **Minji Kim**, S. Kang, K. Joo, H. Li, and P. Kim, "San francisco world: Leveraging structural regularities of slope for 3-DoF visual compass," *IEEE Robotics and Automation Letters (RA-L)*, 2024.

Under Review Paper

- 📖 **Minji Kim**, S.Kang, J.Min, J.Kim, C.Brian, S.Ryan, J.Chen, H.Zhang, P.Kim, "ConicLoc: Map-Free Monocular Localization via Perspective-Two-Circle for Free Flying Robot", *IEEE Robotics and Automation Letters(RA-L)*, 2026
- 📖 J.Han, **Minji Kim**, J.Son, J.Park, M.Wuest, P.Kim, "GALoc: Null-Space Gravity Alignment for Depth-Free Floorplan Localization", *The European Conference on Computer Vision (ECCV)*, 2026

Awards & Grants

- 2024.10  **International Research Fellowship Scholarship,**
Gwangju Institute of Science and Technology.
- 2023.12  **Capstone Design Grand Prize (1st Place),**
Sookmyung Women's University.
- 2023.06/12  **Academic Self-Directed Career Design Award (3rd Place),**
Sookmyung Women's University.
- 2022-2023  **Academic Excellence Scholarship,**
Sookmyung Women's University.

Skills

- Languages  English (fluent), Korean (native)
- Programming Languages  Python, C++, MATLAB
- Frameworks & Tools  ROS2, Docker, VS Code, $\LaTeX$